

LANDR

A working prototype for fatigue-aware ACL injury-risk screening

From a single phone video to a plain-language risk report.

This document presents what the LANDR prototype already does, how it works, the results it produces, what is genuinely novel about it, and how its accuracy will be validated — a buildable, runnable system, not just a concept.

1. Executive Summary

LANDR is a working prototype that screens athletes for ACL injury risk from an ordinary phone video of a jump landing. It reconstructs the athlete's landing mechanics with computer vision, scores their movement against an automated version of the clinical **Landing Error Scoring System (LESS)**, and produces a plain-language risk report and prevention plan. It runs end-to-end today — on synthetic data with no setup, and on real video with a free pose model.

It is built around two focused, defensible contributions identified after a prior-art review: **(1) accuracy** — a temporal + anatomical refinement pipeline plus optional two-view (front+side) capture that targets the single-camera error problem; and **(2) fatigue** — a personal *fatigue-vulnerability* score that measures how an athlete's mechanics degrade when tired, the risk state that actually causes injuries but that standard screening ignores.

4-stage video→report pipeline	19°→7.8° benchmarked valgus error drop*	0→100 fatigue-vulnerability score	26 automated tests passing
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*Simulation of jitter reduction; real-world accuracy requires motion-capture validation (Section 6).

2. The Problem

Around **200,000 ACL injuries** occur in the US each year (~\$7B in cost¹), most of them non-contact — the knee collapses inward on landing. This is visible and measurable before injury. But the accurate way to measure it (a six-figure motion-capture lab) is inaccessible, and the field-friendly alternative (a clinician hand-scoring the LESS) is slow, subjective, and of contested predictive validity.² Most young athletes — especially adolescent girls, who tear ACLs at far higher rates — are never screened, and screening that does happen is done *fresh*, not fatigued.

3. What the Prototype Does

A coach or athlete films a drop-vertical jump on a phone. LANDR then runs four stages:

Stage	Function
1. Perception	Computer vision extracts 3D body landmarks every frame (MediaPipe Tasks pose model).
2. Biomechanics	Computes knee valgus, knee/hip flexion, trunk lean, inter-limb asymmetry, and detects the landing events (contact, lowest point).
3. Scoring	Automated, auditable LESS + a 0–100 risk score (each item shows its measured value and threshold).
4. Reasoning	A vision-language model (or an offline templated generator) writes a plain-language risk report + corrective drills.

Two capability modules sit on top:

- **Fatigue (Pillar 2):** compare the same athlete fresh vs. fatigued and output a personal fatigue-vulnerability score — within-athlete change that cancels most measurement bias.
- **Accuracy (Pillar 1):** a refinement pass (temporal smoothing + anatomical limb-length constraint), a stronger pose model option, and two-view (front+side) fusion that takes valgus from the front and knee/trunk flexion from the side.

4. How It Works (Architecture & Stack)

The system is modular — each stage has a clean data contract, so any stage can be upgraded or validated independently.

```
video ■■ pose ■■ biomechanics ■■ scoring ■■ reasoning ■■ risk report
(MediaPipe) (angles+events) (auto-LESS) (VLM / offline)

accuracy.py : temporal smoothing + anatomical constraint (refine)
multiview.py : fuse front (valgus) + side (flexion) clips
fatigue.py : fresh vs fatigued -> vulnerability score
```

Component	Technology	Notes
Pose	MediaPipe Tasks (lite/full/heavy)	Pre-trained; upgrade path to Sports2D / OpenCap / Pose2Sim.
Biomechanics	NumPy / SciPy	Pure, unit-tested geometry — no training needed.
Scoring	Rule-based (transparent)	Explainable baseline; thresholds centralized & tunable.
Reasoning	GPT-4o / Gemini / offline mock	Pluggable; graceful fallback, no key required.
Delivery	CLI + Colab notebook	Runs with no install via synthetic demo.

Importantly, **nothing requires training**: the vision model is pre-trained, and the rest is rule-based math.

5. Prototype Results (What It Actually Outputs)

All figures below are real outputs from the current prototype.

Basic screening (synthetic landings)

Input	LESS total	Risk	Risk score
Aligned, soft landing ('good')	1	LOW	13.3 / 100
Stiff, valgus landing ('poor')	8	HIGH	100 / 100

Fatigue-vulnerability (Pillar 2 demo)

Same athlete, fresh vs. fatigued: peak knee valgus increased **+24.6°**, knee flexion at contact dropped, automated LESS rose by 5 — yielding a fatigue-vulnerability score of **100/100 ('vulnerable')**. The fresh-vs-fatigued overlay plot clearly shows the knees staying aligned when fresh (~5°) and collapsing (~20–30°) when fatigued.

Accuracy benchmark (Pillar 1 demo)

Metric	Naive RMSE	Refined RMSE	Reduction
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Knee valgus	19.0°	7.8°	~59%
Knee flexion	21.4°	9.3°	~56%

Simulation of jitter reduction on synthetic data; demonstrates the method, not real-world accuracy (see Section 6). The naive baseline matches the ~18–20° single-camera valgus error reported in the literature.³

6. What Is Novel, and How Accuracy Is Validated

Novelty (vs. prior art)

Automated LESS and single-camera valgus already exist (OpenLESS; MediaPipe studies).^{3,4} LANDR's contribution is the combination the prior work leaves open: an **accessible, fatigue-aware, longitudinal** landing screen, with (a) a measured accuracy gain and (b) a per-athlete fatigue-vulnerability signature.

Validation plan

Claim	Ground truth	Metric → target
Accuracy gain	Marker mocap + force plate (public sets)	valgus/flexion RMSE → toward <10°
Fatigue detection	Fresh vs. fatigued dataset ⁵	Δ valgus/flexion; classifier AUC
Whole system	Human-LESS baseline	added value over a fresh static screen

The next milestone is running the pipeline against a public motion-capture dataset to report a **real** error number — the step that turns 'working demo' into 'validated tool.'

7. Status, Limitations & Roadmap

Status: end-to-end runnable (synthetic + real phone video), one-click Colab notebook, 26 passing automated tests, two-view + fatigue + benchmark all implemented.

Honest limitations:

- Single-camera absolute accuracy is limited by depth; mitigated by two-view capture and within-athlete fatigue comparison.
- The automated LESS is an approximation with contested predictive validity — used as a baseline to improve upon.
- The accuracy benchmark is a simulation; real validation against mocap is the next step.
- Outputs are decision-support, not a diagnosis; a clinician stays in the loop.

Roadmap: validate on public mocap data → pilot with one team/clinic (consented data) → app + learned, outcome-validated risk model → longitudinal season tracking.

8. How To Run It

```

pip install numpy scipy matplotlib
python -m landr.cli demo --quality poor # basic screen
python -m landr.cli fatigue --plot f.png # fatigue-vulnerability
python -m landr.cli benchmark # accuracy gain
# real video (needs mediapipe):
python -m landr.cli analyze jump.mp4 --plot out.png
python -m landr.cli analyze2 front.mp4 side.mp4 --plot out.png # best accuracy

```

References

1. ACL injuries & cost (~200k/yr, ~\$7B), Panorama Orthopedics.
<https://www.panoramaortho.com/resource/acl-injuries-by-the-numbers/>
2. LESS validity (JUMP-ACL; and contested prospective evidence), PubMed. <https://pubmed.ncbi.nlm.nih.gov/19726623/>
3. Knee valgus via MediaPipe pose estimation (~18.8–19.7° error), Heliyon 2024.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11399566/>
4. OpenLESS: automated Landing Error Scoring System, medRxiv 2024.
<https://www.medrxiv.org/content/10.1101/2024.11.28.24318160v1.full>
5. Fatigued vs non-fatigued jump-landing motion-capture dataset, Nature Scientific Data 2025.
<https://www.nature.com/articles/s41597-025-05934-5>

Prototype results are real outputs of the current LANDR codebase. Accuracy benchmark is a simulation of jitter reduction; real-world accuracy requires motion-capture validation. Decision-support only — not a diagnosis.